Lab Number : 32 Date:2026/03/19

Title: Configuring SSH server to allow/deny root login & allow/deny user login.

1.Install required package for Open SSH Server.

Command:

```
root@localhost:~# sudo yum install openssh-server -y
CentOS Stream 10 - BaseOS                       3.6 kB/s | 5.3 kB     00:01
CentOS Stream 10 - AppStream                    6.1 kB/s | 5.4 kB     00:00
CentOS Stream 10 - Extras packages              2.3 kB/s | 5.7 kB     00:02
Package openssh-server-9.9p1-20.el10.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package                Architecture  Version                Repository     Size
================================================================================
Upgrading:
 openssh                x86_64        9.9p1-21.el10          baseos        351 k
 openssh-clients        x86_64        9.9p1-21.el10          baseos        766 k
 openssh-server         x86_64        9.9p1-21.el10          baseos        548 k

Transaction Summary
================================================================================
Upgrade  3 Packages
```

Output:

```
root@localhost:~# rpm -qa | grep openssh
openssh-9.9p1-21.el10.x86_64
openssh-clients-9.9p1-21.el10.x86_64
openssh-server-9.9p1-21.el10.x86_64
root@localhost:~#
```

2. Allow ssh packets to enter through the firewall.

Command:

```
root@localhost:~# sudo firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
root@localhost:~# sudo firewall-cmd --reload
success
root@localhost:~# sudo firewall-cmd --list-services
cockpit dhcpv6-client http ssh
```

3. Start and enable ssh service.

Command:

```
root@localhost:~# sudo systemctl start sshd
root@localhost:~# sudo systemctl enable sshd
root@localhost:~# sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-03-19 07:05:59 +0545; 8min ago
 Invocation: aa4a789d488d4586829bda1f11cbdcce
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 4470 (sshd)
      Tasks: 1 (limit: 20120)
     Memory: 1.3M (peak: 1.6M)
        CPU: 20ms
     CGroup: /system.slice/sshd.service
             └─4470 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 19 07:05:59 localhost.localdomain systemd[1]: Starting sshd.service - OpenS>
Mar 19 07:05:59 localhost.localdomain sshd[4470]: Server listening on 0.0.0.0 p>
Mar 19 07:05:59 localhost.localdomain sshd[4470]: Server listening on :: port 2>
Mar 19 07:05:59 localhost.localdomain systemd[1]: Started sshd.service - OpenSS>
lines 1-17/17 (END)
```

4. Configure SSH Server to deny direct root login.

Command:

```
root@localhost:~# sudo vi /etc/ssh/sshd_config
```

Finding and changing the PermitRootLogin : no

```
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
-- INSERT --                                                    40,20         30%
```

Restart SSH Server:

```
root@localhost:/# sudo systemctl restart sshd
```

5. Configure OpenSSH Server to block login from Users.

Command:

```
root@localhost:/# sudo systemctl restart sshd
```

Adding to bottom:

```
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# override default of no subsystems
Subsystem       sftp    /usr/libexec/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs serve
AllowUsers student
AllowUsers student@192.168.0.1*
DenyUsers production
```

Restart Server:

```
root@localhost:/# sudo systemctl restart sshd
```

Output:

```
root@localhost:/# sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-03-19 07:37:36 +0545; 11s ago
 Invocation: 2fc250ac232f4ecea0432f013f216b79
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 8122 (sshd)
      Tasks: 1 (limit: 20120)
     Memory: 1.2M (peak: 1.5M)
        CPU: 20ms
     CGroup: /system.slice/sshd.service
             └─8122 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 19 07:37:36 localhost.localdomain systemd[1]: Starting sshd.service - OpenS>
Mar 19 07:37:36 localhost.localdomain sshd[8122]: Server listening on 0.0.0.0 p>
Mar 19 07:37:36 localhost.localdomain sshd[8122]: Server listening on :: port 2>
Mar 19 07:37:36 localhost.localdomain systemd[1]: Started sshd.service - OpenSS>
lines 1-17/17 (END)
```